



SYMPOSIUM ARTICLE

How Damage, Recovery, and Repair Alter the Fitness Impacts of Thermal Stress

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Synopsis The fitness implications of climate variability and change are often estimated by integrating an organism’s thermal sensitivity of performance across a time series of experienced body temperatures. Although this approach is an important first step in evaluating an organism’s sensitivity to climate or climate change, it ignores potential influences of recent exposure to thermal stress on current thermal sensitivity. Here, we account for recent thermal stress by estimating rates of damage, repair, and other carryover effects; and we illustrate the approach with fecundity and development rate data from experiments that exposed aphids to various stressful and fluctuating temperatures. Our analyses indicate that heat stress for these aphids starts near the upper thermal limit for performance; that heat stress intensifies with both the exposure duration and with temperature; and that there is considerable capacity for repair at temperatures near the thermal optimum for performance. Results from experiments with aphids indicate that incorporating time series of damage, recovery, and repair will be necessary to anticipate fitness outcomes of climate change and variability.

Introduction

Better quantification of environmental exposure and of organismal sensitivity is improving projections of biological responses to variable and changing environments (Buckley and Kingsolver 2021). However, an ongoing challenge involves accurately characterizing organismal sensitivity. Traditionally, organismal sensitivity has been quantified by using thermal performance curves (TPCs), which describe the rates of fitness-conferring biological processes as a function of body temperature, T_b (Huey and Stevenson 1979). TPCs are typically unimodal and skewed: performance increases gradually to a thermal optimum (T_{opt}) but then declines rapidly with stressful temperatures to zero performance (at T_{max} , often assessed as a critical thermal maximum, Fig. 1a). TPCs can be generalized to other physiological traits such as hydration or pH status, but here, we refer to body temperatures for simplicity (Buckley and Kingsolver 2021). TPCs can be integrated across T_b distributions to estimate aggregate performance over

some time period (Huey and Slatkin 1976; Deutsch et al. 2008; Vasseur et al. 2014).

An organism’s thermal history can result in persistent changes in thermal sensitivity that violate the integration approach (above), which assumes aggregate performance is independent of the sequence and levels of prior T_b (Williams et al. 2016). Sinclair et al. (2016) highlight other assumptions that can be particularly problematic in thermally variable environments. For example, studies typically measure a specific performance metric (e.g., growth, sprint speed) and implicitly assume that this metric characterizes overall organismal performance or even fitness (Huey 1982). Also, TPCs are often assumed to be invariant, even though TPCs often vary between performance traits (Angilletta 2009; Kellermann et al. 2019), fitness indices (Huey and Berrigan 2001; Williams et al. 2016), and life stages (Kingsolver et al. 2011; Kingsolver and Buckley 2020).

TPCs are usually derived from experiments that first acclimate individuals to constant and benign

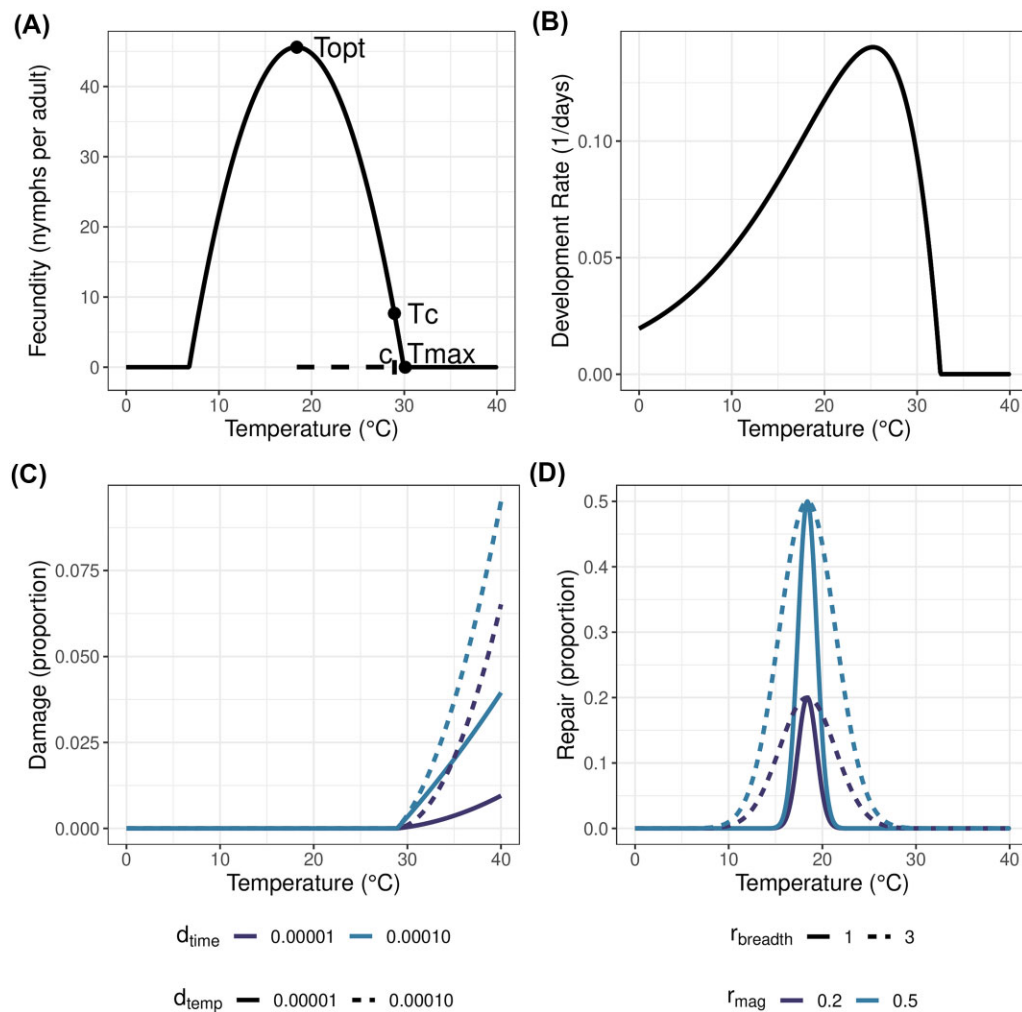


Fig. 1 We illustrate the constant temperature TPCs for (A) fecundity and (B) development rate. A critical temperature threshold, T_c , indicates the onset of stress and is estimated using a parameter c , which is the proportional distance between thermal optimum T_{opt} and thermal maximum T_{max} at which stress initiates. We also depict how parameter values influence rates of (C) damage and (D) repair for a single time point (the third hour of exposure to a stressful temperature). (C) We assume that damage accumulates with increasing exposure time (d_{time} , line color) and temperature (d_{temp} , line type) above T_c (depicted with the assumption that $c = 0.9$). (D) We assume repair follows a Gaussian function centered at a thermal optimum for repair T_r (depicted assuming $T_r = T_{opt}$) and fit the height (r_{mag} , color) and breadth ($r_{breadth}$, line type).

temperatures. Next, performance of some traits (e.g., sprint speed of lizards, heart rate of crabs) is measured immediately after T_b is shifted to multiple temperatures, but that of other traits (e.g., population growth rates of insects) require chronic exposures of individuals to a single temperature. For the former, TPCs can be measured for individuals; but for the latter, TPCs are often based on pooling data across individuals.

Accurately aggregating performance over time requires aligning the time scale of measurement to the timescales at which organisms respond to environmental variation (Colinet et al. 2015; Buckley et al. 2023). The non-linearity of TPCs and other measures of thermal sensitivity results in deviations between performance at mean temperatures and the mean of

performance integrated across temperatures (Jensen's Inequality) (Martin and Huey 2008; Denny 2017). Temperature fluctuations restricted to below an organism's T_{opt} increase mean performance, but fluctuations at higher temperatures reduce mean performance (Kingsolver et al. 2015; Kroeker et al. 2020). A synthesis of experiments found that increases in mean temperatures, greater variability, and past exposures to thermal variability accentuate the extent to which thermal variability in the stressful temperature range detriments performance (Slein et al. 2023). Here, we examine responses to past thermal stress, which we define as physiological shifts, declines in performance, and damage accumulation in response to low or high temperatures.

Although some studies indicate that non-linear averaging techniques can use TPCs measured at fixed T_b to effectively estimate responses to variable environments (Bernhardt et al. 2018), others find substantial deviations: carryover effects generated by stress, acclimation, and damage can alter TPCs (Dillon et al. 2007; Kingsolver et al. 2015; Marshall et al. 2021). In particular, exposures to high T_b can have damaging effects on performance, and even brief exposures can reduce performance—at least transiently (Kingsolver and Woods 2016; Ma et al. 2021). Low temperatures are increasingly stressing organisms due to contemporary increases in environmental variability (Marshall et al. 2020). Studies of physiological sensitivity to low temperatures provide complementary insights (Williams et al. 2014), but we focus here on responses to high temperatures because most organisms spend more time closer to stressfully hot temperatures (table II in Huey 1982) and, given the skewed shape of TPCs, will do so increasingly with climate warming (Colinet et al. 2015).

The magnitude, duration, and frequency of exposure to stressful T_b s can alter the shapes of TPCs transiently. Brief exposure to stressful temperatures could reduce performance over a wide or a narrow range of T_b s. Presumably, the magnitude of such decrements increases with the magnitude of T_b above the temperature threshold corresponding to the onset of stress (Jørgensen et al. 2021). The decrement in performance tends to be an accelerating function of the temperature deviation above the threshold and of exposure-time as stress and damage compound (Ørsted et al. 2024). Of course, organisms experience a temperature sequence, so intermediate T_b matter. Moreover, past exposures to stressful temperatures can have carryover effects and thus decrease (“heat hardening”) (Loeschcke and Hoffmann 2007; Baeza Icaza et al. 2025) or increase sensitivity to subsequent exposures (Williams et al. 2016).

An initial and necessary step toward better quantifying how variable environments alter organismal sensitivity involves accounting for how past exposure to thermal stress influences current performance. Our analysis builds on several models that describe the performance implications of heat stress. Kingsolver and Woods (2016) described temporal dynamics of gene expression and production of heat shock proteins. They modeled—and empirically validated—how the cost of producing stress proteins lowers subsequent growth rate. Klanjscek et al. (2016) developed a damage and repair model for oxidative stress, which is a primary mechanism of thermal damage. They find that past exposures can substantially augment stress. We take a more descriptive approach to assessing damage and repair dynamics.

A new generation of experiments examining biological responses to environmental variability affords the opportunity to assess empirically how recent thermal history shapes current thermal performance. In mussels, thermal fluctuations at cool mean temperatures accelerate growth relative to constant temperatures, those at intermediate mean temperatures slow growth due to stressful peak temperatures, and those at hot mean temperatures accelerate growth by providing relief (Vajedsamiei et al. 2021; Gerhard et al. 2023). Both the lifecycle timing of and number of heat exposures influences the magnitude of growth rate declines of *Manduca sexta* larvae (Kingsolver et al. 2021).

Here, we synthesize past research on damage resulting from encountering acute thermal stress; responses to repeated exposures, including the potential for acclimation or hardening; and repair trajectories and their dependence on recovery temperatures. We build on this past research to consider how thermal history affects fitness implications in fluctuating environments. Then, we leverage an extensive series of experiments by Ma and colleagues who exposed *Sitobion avenae* (English grain aphids) to multiple stressful and fluctuating temperature regimes (Zhao et al. 2014, 2019; Ma et al. 2015a, 2018). We fit a function to the aphid data to gain qualitative understanding of how thermal stress, damage, and repair interact over time. We hypothesize that acute thermal stress and ultimately damage accumulate with the magnitude of and exposure time to stressful temperatures but can be partially offset by recovery if conditions are permissive. We then discuss how our initial attempt to account for thermal history can inform coupled empirical and modeling approaches to better account for the impacts of thermal history on organismal sensitivity to variable environments.

Damage

Several approaches have previously been developed to quantify damage. Damage associated with misfolded or unfolded proteins, oxidation, and osmotic or ionic imbalance is a primary physiological mechanism by which thermal stress alters TPCs, but few studies characterize the molecular or physiological mechanisms underlying such damage (González-Tokman et al. 2020). Thermodynamic approaches describe the exponentially increasing rates of death and damage with increasing temperatures (Pearce 2013). Arrhenius models estimate the number of molecules with kinetic energies exceeding a biochemical reaction's activation energy and are thus reactive, and these models provide a thermodynamic explanation for exponential rate increases in damage at high T_b (Dell et al. 2011).

The thermal death time (TDT) framework focuses on death not damage, but likewise assumes that survival declines exponentially with increasing temperature and exposure time (Rezende et al. 2014). A related deterministic model explicitly incorporated damage accumulation to account for the rapid exponential decline of survival at stressfully high temperatures (Jørgensen et al. 2021). An extension to this framework distinguishes between permissive and stressful temperature ranges (separated by a “critical temperature” T_c), with different exponential scaling for each range (Jørgensen et al. 2021, 2022). Further work extended the exponential temperature dependence of survival to reproductive output (Ørsted et al. 2024).

Stress can sometimes be repeated over time. Responses to multiple cold stress exposures has been perhaps better documented than such responses to heat stress (Colinet et al. 2015). Interspersing cold exposures with non-stressful temperatures tends to reduce mortality (Marshall and Sinclair 2010), via mechanisms such as allowing the reallocation of energy stores (Marshall and Sinclair 2012). Responses to multiple exposures can vary among performance metrics and also depend on the extent of prior thermal acclimation/hardening (Kingsolver et al. 2021). Temporal dynamics of the gene expression and the resultant production of heat shock and other proteins are key determinants of magnitudes and timescales of acclimation (Sørensen et al. 2003). However, the timescales of heat shock protein expression and down regulation tend to be substantially slower than the timescales of acute thermal damage (Feder and Hofmann 1999).

Recovery and repair

Some approaches to quantifying performance in fluctuating environments assume that the damage induced by thermal stress during the day can be repaired overnight. Such an assumption yielded reasonable predictions of *Drosophila* survival probabilities in both the laboratory and field (Rezende et al. 2020). But other physiological studies suggests that recovery takes longer than overnight (Colinet et al. 2015). Repair seems to be fastest at body temperatures near the thermal optimum for performance (MacFadyen et al. 2004; Morison et al. 2020). For example, *Drosophila* recovery from heat knockdown is fastest near T_{opt} (Ørsted et al. 2022). Life histories such as reproductive status can influence the nature and extent of damage and the potential for recovery and repair (Aprison and Ruvinsky 2014; Sales et al. 2021).

Methods

System and parameterization

We leverage a series of experiments by Ma et al. who systematically measured the responses on English grain aphids, *Sitobion avenae*, to a variety of thermal stresses (Zhao et al. 2014, 2019; Ma et al. 2015a, 2018). In most experiments (see the papers for details), newborn nymphs of recently field-collected aphids were placed individually in cages that were attached to leaves of wheat seedlings. Aphids were reared under sawtooth or sine-wave diurnal temperature fluctuations. These aphids are live bearing and were checked 1–2 times daily to assay fitness components (development, survival, reproduction). We focus on fecundity, which was estimated as the lifetime total number of nymph offspring per adult, and on development rate (1/days) of the nymph stage. Our analyses assumes that energy assimilation and other fitness enhancing processes aggregate over time to determine total fecundity. We do not analyze two additional metrics (survival rate, adult longevity) recorded by Ma and colleagues, as these metrics were not measured in all experiments.

Under constant thermal conditions, prior research (Zhao et al. 2014) has described aphid fecundity (nymphs per adult) as a polynomial function of temperature T ($c_1 \times c_2 T \times c_3 T^2$, where $c_1 = -69.1$, $c_2 = 12.49$, and $c_3 = -0.34$) (Fig. 1A). This TPC has a T_{opt} of 18.5°C and a T_{max} of 30.1°C. Nymphal development rate (1/days) has been described (Ma et al. 2015b) as an exponential function of temperature T ($e^{c_1 T} - e^{c_2 - (T_{max} - T)/c_3}$, where $T_{max} = 32.947$, $c_1 = 0.137$, $c_2 = 4.514$, and $c_3 = 7.267$) (Fig. 1B).

To examine how damage and repair influence fitness outcomes, we use data from seven experiments that exposed the aphids to diverse temperature regimes (Fig. 2). All experiments exposed aphids to temperatures above T_{max} . [Recall, the fecundity T_{opt} for these aphids is 18.05°C and T_{max} is 30.1°C.]

- (1) Experiment 1 studied how various nighttime temperatures (including minima above or below T_{opt}) influenced rates of repair (Fig. 2A). Daytime temperatures in all treatments diverged symmetrically from 27°C (at 9 and 21 h) and peaked at 35°C at 15 h, but nighttime minimum temperatures dropped linearly to 13, 16, 19, 21, 23, or 25°C (Zhao et al. 2014).
- (2) Experiment 2 studied how different daytime maximum temperatures (32, 34, 36, 38, and 40°C)—but constant nighttime temperatures—influenced development and fecundity (Fig. 2B). Nighttime temperatures were held constant at 22°C (Ma et al.

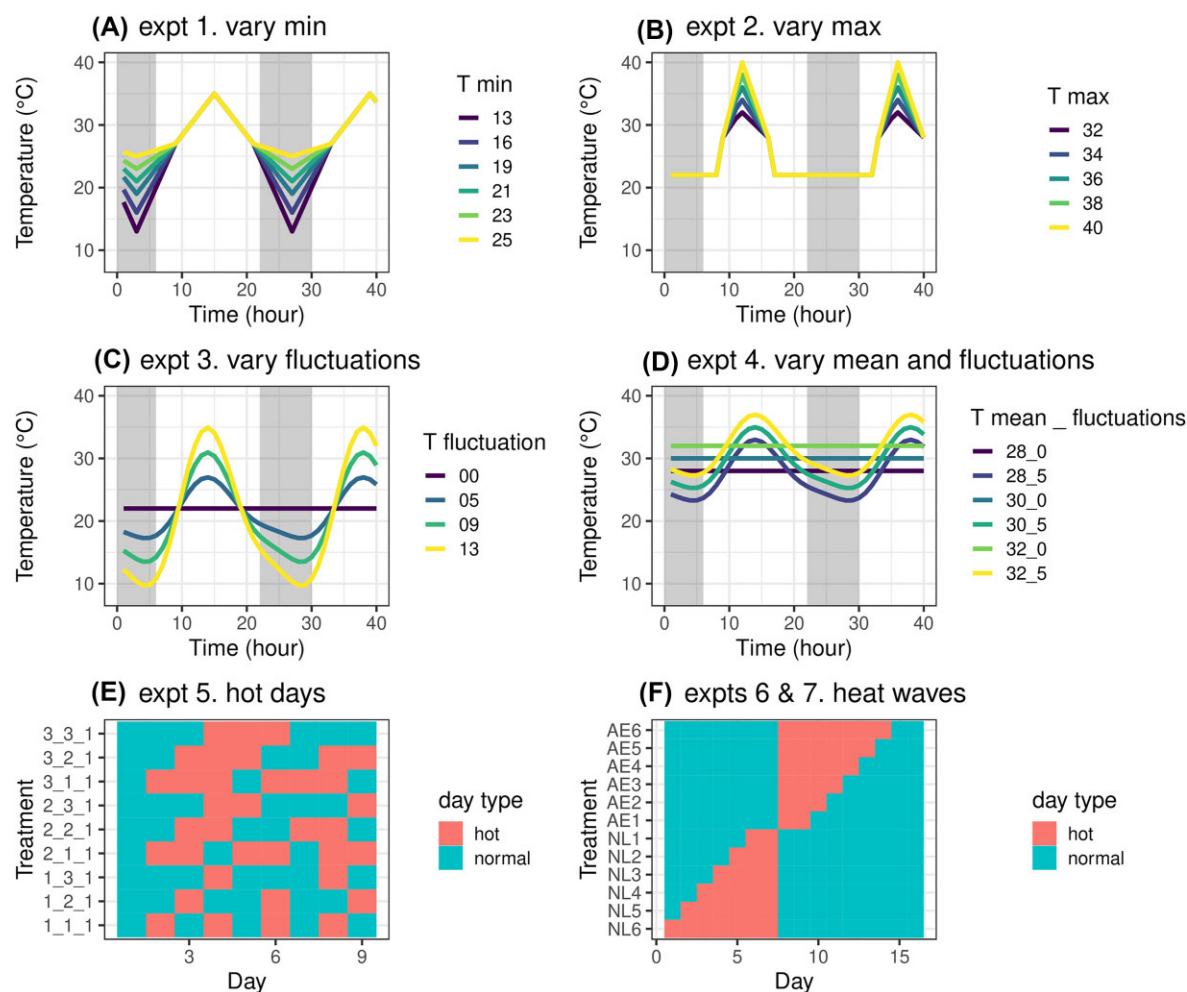


Fig. 2 The temperature time series for experiments that vary daily minima (expt 1, Fig. 2A), daily maxima (expt 2, Fig. 2B), temperature variation (expt 3, Fig. 2C), and high thermal means and variation (expt 4, Fig. 2D). Aphids experienced 8 hours of darkness daily (gray shading for expts 1–4). Additional experiments (Fig. 2E, F) simulate heatwaves by interspersing normal (13–28°C) and hot (20–35°C) days. Expt 5 examined responses to varying the number of normal and hot days between 1 and 3; the treatment names indicate the number hot days, number normal days, and whether the treatment started with a normal (1) or hot (2) day. We focus on results starting on normal days. Additional experiments examined responses to heatwaves during the adult (expt 6) and nymphal (expt 7) stage that varied in duration between 1 and 6 days.

2015a). We dropped several outlier aphids with exceptionally fast development rate.

- (3) Experiment 3 monitored the impact of daily fluctuations of various magnitudes (0, 5, 9, and 13°C) around a mild (but above T_{opt}) mean temperature of 22°C (Fig. 2C) (Wang and Ma 2023).
- (4) Experiment 4 studied the impact of varying the magnitude of symmetrical daily fluctuations (0 and 5°C) around a series of high—potentially stressful—mean temperatures (28, 30, and 32°C) (Fig. 2D) (Wang and Ma 2023).
- (5) Experiment 5 studied the impact of alternating periods of normal and of hot days, where the sequence initiated on either a normal or hot day (Fig. 2E) (Ma et al. 2018). We focus analysis on results starting on hot days. The experiment implemented all combinations of 1 to 3 normal days, which fluctuated between 13°C and 28°C, followed by 1 to 3 hot days, which fluctuated between 20°C and 35°C. This experiment explores the impact of heat waves and recovery periods.
- (6) Experiment 6 examined responses of adults to heatwaves of various durations (1 to 6 days) (Fig. 2F). Exposure to heat waves initiated at the start of the adult period. Normal- and hot-day temperatures were as in experiment 5 (Zhao et al. 2019). (Development rate data were not available for this experiment.)
- (7) Experiment 7 corresponds to Experiment 6 but monitored responses of nymphs to heatwaves ranging from 1 to 6 days that concluded at the end of the nymphal period (Fig. 2F).

Damage and repair function

In the various experiments (above), aphids were exposed to diverse temperature regimes and then scored for development rate as nymphs and cumulative fecundity as adults. From these observed patterns, we use a mathematical function (below) that enables us to evaluate the dynamics of damage and of repair in response to fluctuating temperatures, encompassing thermally stressful periods. We use a simple function because our ability to resolve details is limited by our fitting performance estimates to outcomes at the end of the experiments (rather than continuous performance measures). We estimate performance using an hourly time-step, which is appropriate for the smooth, diurnal fluctuations examined here but might be inadequate to capture responses to acute thermal stress. We fit a parameter c constrained between 0 and 1 to estimate where damage initiates between the optimum and endpoint of performance (Fig. 1A). We estimate the critical temperature corresponding to the onset of damage as $T_c = T_{\text{opt}} + c \times (T_{\text{max}} - T_{\text{opt}})$. We assume that damage in response to temperature T accumulates as a function of both exposure time (t) above and thermal distance above T_c :

$$\text{damage} = d_{\text{time}}t + d_{\text{temp}}\max(T - T_c, 0),$$

where d_{time} and d_{temp} are coefficients describing the rate of responses to increased exposure time and temperature, respectively (Fig. 1C). We expect accelerating (e.g., exponential) responses to exposure time and temperature. However, initial model exploration revealed an inability to distinguish linear and exponential responses given our coarse response metric of experiment fitness outcomes. We thus chose to simplify the model by assuming linear responses. We sum damage over time and constrained cumulative damage between 0 and 1.

To estimate repair (which we consider a component of recovery), we use a Gaussian function with a height of 1 centered at T_r , the temperature corresponding to the maximum repair rate (Ørsted et al. 2022). Parameter r_{mag} varies between 0 and 1 and describes the maximum proportional repair during a time interval. Parameter r_{breadth} describes the breadth of the Gaussian function and thus how quickly repair rate declines away from T_r (Fig. 1D).

For each hourly time period, cumulative damage is discounted by a rate of repair:

$$\begin{aligned} \text{repair rate} &= r_{\text{mag}} \times \exp(-(T - T_r)^2 / (2r_{\text{breadth}}^2)) \\ \text{cumulative damage} &= \text{cumulative damage} \\ &\times (1 - \text{repair rate}). \end{aligned}$$

We used the TPCs estimated under constant conditions (above) to estimate performance (p) in the absence of acute thermal stress. This approach suffers from

timescale mismatches (applying chronic temperature responses to assessing responses to short term temperature variation) but is a standard and feasible approach to estimating responses to temperature variation. We note the TPC decline in performance in response to sustained high temperatures is attributable to thermal stress (Kingsolver and Woods 2016), which leads to our analysis potentially overestimating and double counting stress responses between T_{opt} and T_{max} . We discount performance estimates for each hour multiplicatively to account for damage and repair:

$$p \times (1 - \text{cumulative damage}).$$

We estimate performance over the multi-week experiments as the mean p across hours. We compared these average performance estimates to observations of life-time fecundity and developmental rate for each experiment. We started averaging performance after 4 days to minimize the influence of early transient dynamics. We average over a period that is longer than either nymphal development or the reproductive period, but the extended average should not qualitatively influence results due to the repetition of thermal fluctuations in the experiment (excepting experiments 6 and 7). For cases where observed performance outcomes exceed the estimates of performance assuming no damage (due to different populations, using TPCs measured under chronic thermal conditions, or beneficial responses to thermal fluctuations), we scaled the estimated performance to the maximum (across treatments) ratio of observed to estimated performance.

We depict how parameters influence damage and repair dynamics over time in Fig. 3. Higher values of the coefficients corresponding to rates of damage due to exposure time (d_{time}) and to temperature above T_c (d_{temp}) decrease fitness estimates, and the damage accumulates through time. A higher maximum repair rate (r_{mag}) and a broader temperature breadth for repair (r_{breadth}) restores performance following thermal stress.

We apply this modeling framework to address two primary questions:

- (i) How does damage depend on the magnitude and duration of thermal stress? We use the parameter c to assess where the TPC damage initiates. Estimates of d_{temp} and d_{time} indicate the relative impacts of exposure temperature and time, respectively.
- (ii) What is the potential for recovery and repair, and how does it depend on temperature? Estimates of r_{mag} and r_{breadth} indicate the proportional magnitude of repair and its dependence on temperatures near the thermal optimum for repair, respectively.

We use the experimental fitness component outcomes to estimate optimal parameter values for each

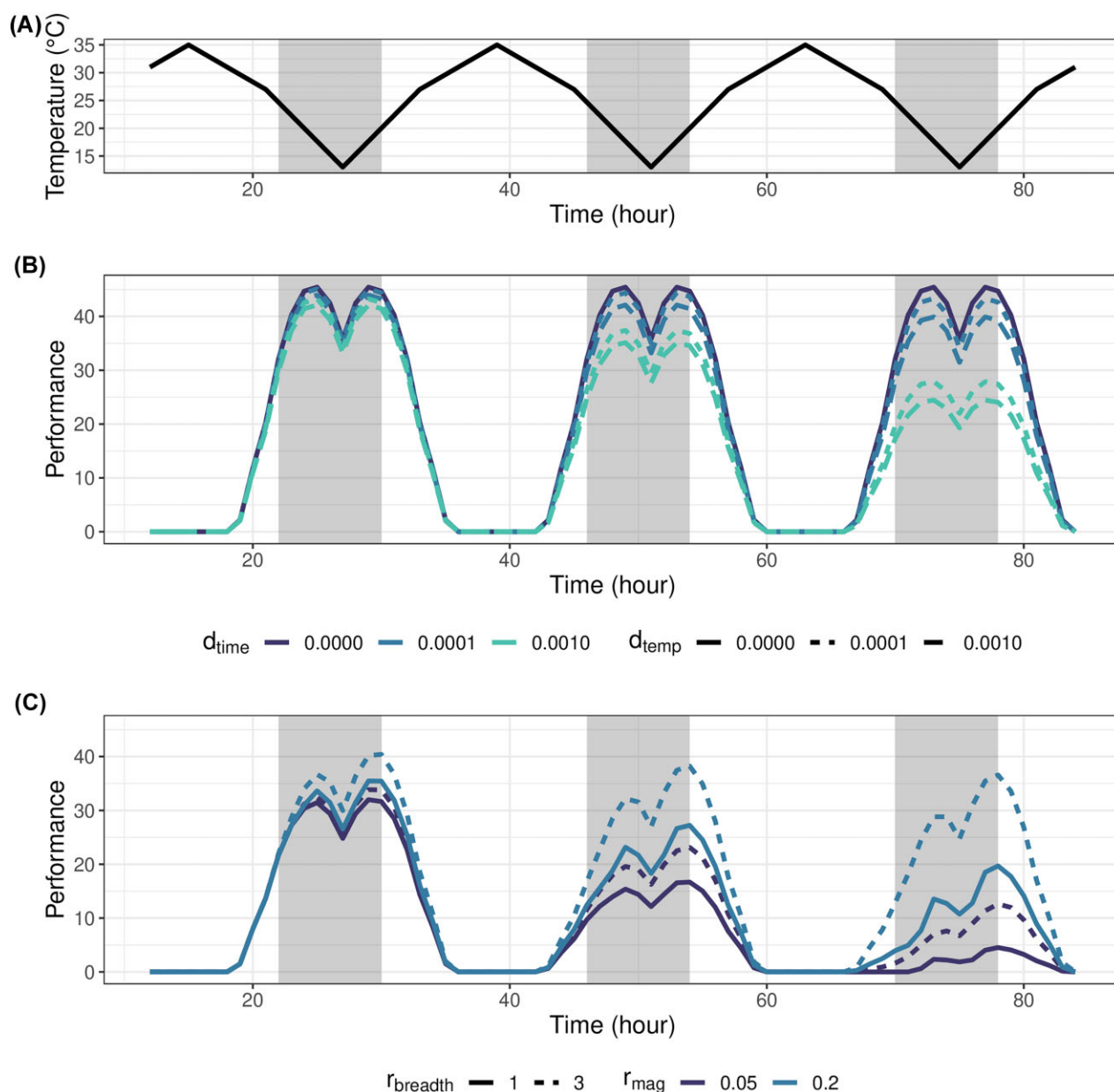


Fig. 3 The temporal dynamics of damage and repair impacting rates of performance that determine fecundity (nymphs per adult). Temperature fluctuations (A) result in a time series of damage (B) that depends on exposure time (d_{time} , line color) and temperature (d_{temp} , line type) over a critical temperature threshold. (C) For assumed damage dynamics ($d_{\text{time}} = 0.001$, $d_{\text{temp}} = 0.01$), the magnitude (r_{mag} , line color) and thermal breadth (r_{breadth} , line type) of the repair function alters outcomes. Night hours are shaded.

experiment. We focus on fitting four key parameters: d_{time} , d_{temp} , r_{mag} , and r_{breadth} (scenario 1, Table 1). We assume, based on exploratory fitting, that $c = 0.9$ such that $T_c = T_{\text{opt}} + 0.9 (T_{\text{max}} - T_{\text{opt}})$ (see Table S1 scenario 2). We set T_r as the T_{opt} for performance as a default. We subsequently implemented a series of scenarios to assess these assumptions as well as statistical support for accounting for damage and repair (Table 1). Scenario 2 assesses the assumption that damage onsets near T_{max} ($c = 0.9$) by additionally estimating the c parameter. Scenario 3 assesses the assumption that the ther-

mal optimum for repair equals the thermal optimum for performance ($T_r = T_{\text{opt}}$) by additionally estimating the T_r parameter. The final three scenarios assess support for model components by comparing models with and without their inclusion. Scenario 4 drops the exposure time component of damage ($d_{\text{time}} = 0$). Scenario 5 drops the exposure temperature component of damage ($d_{\text{temp}} = 0$). Scenario 6 drops repair from the model ($r_{\text{mag}} = 0$).

We implemented the function and fitting in R. We used the R function `nmkb` in the R `dfoptim` package

Table 1 The scenarios used to estimate parameter value and support for accounting for damage and repair.

Parameter	Description	Scenario					
		1	2	3	4	5	6
d_{time}	response rate to exposure time				0		
d_{temp}	response rate to exposure temperature					0	
r_{mag}	maximum proportional repair during a time interval						0
r_{breadth}	breadth of Gaussian repair function						0
c	proportional temperature of damage onset between T_{opt} and T_{max}	0.9		0.9	0.9	0.9	0.9
T_r	thermal optima for repair	T_{opt}	T_{opt}		T_{opt}	T_{opt}	T_{opt}

We indicate fixed (assumption shown) and fit parameters (gray shading). See text for detailed parameter descriptions.

to implement Nelder-Mead derivative-free optimization to align parameter values with the experimental results subject to box constraints (Varadhan et al. 2023). Parameters were optimized by minimizing the sum of squared error. All optimizations converged by the differential of the best fit exceeding a threshold. We sum error across treatments, weighted by the number of replicates in the treatment. We also estimate Akaike information criterion (AIC) values. We constrained the parameters to yield realistic dynamics: $0 \leq d_{\text{time}} \leq 0.5$; $0 \leq d_{\text{temp}} \leq 0.5$; $0 \leq r_{\text{mag}} \leq 1$; $0 \leq r_{\text{breadth}} \leq 3$; $0 \leq c \leq 1$; and $10 \leq T_r \leq 25$.

Results

Fecundity

Both damage and repair substantially alter fecundity through time (Fig. 4). For each experiment, we summarize the parameter estimates (Table 2), how they vary among scenarios (Table S1), and their implications for fecundity estimates.

In Experiment 1, which varied nighttime minimum temperatures, fecundity declines when nighttime minimum temperatures exceed the T_{opt} for performance (Fig. 4A). Performance estimates predict the observed fecundity declines with increasing nighttime minimum temperatures. However, they miss the abrupt decline to low fecundity for nighttime minimum temperatures above T_{opt} , and this decline may indicate shift in stress mechanisms at high T_b (Fig. 5A). The function estimates a similar impact of exposure time and temperature, but simplifying the model by dropping the exposure time or temperature effect or repair improved the model fit (Table S1). The preference for a simpler model is most pronounced for this experiment, which includes relatively minor thermal stress (Table S1).

In Experiment 2, which varied daytime maximum temperatures, observed fecundity is rather constant with temperature until the daily maximum temperature exceeds 36°C: it drops drastically at higher temperatures (Fig. 5B). The performance function with damage and repair predicts that fecundity drops to zero beyond 32°C. The best fitting function includes low estimates for both the exposure time and temperature components of damage but relatively bigger values for the temperature component (Table 2, Table S1). Dropping either the exposure time or temperature components of damage similarly reduces function performance, suggesting that both forms of compounding similarly stress organisms.

In Experiment 3, which varied the magnitude of fluctuations around a mild mean temperature, both performance estimates miss the observed fecundity reductions with either no or the highest fluctuations (Fig. 5C). We estimate that exposure time contributes to damage rates more than temperature for the primary scenario but not others (Table 2, Table S1). When dropping exposure time or temperature effect or repair, we estimate small performance changes.

In Experiment 4, which varied the magnitude fluctuations around a series of high mean temperatures, the function overestimates the fecundity declines at high mean temperatures (Fig. 5D). We estimate a very small or zero impact of exposure time and a larger temperature effect in the primary scenario but dropping either term does not weaken fit to observations (Table S1). We estimate a high magnitude of repair over a broad thermal breadth.

In Experiment 5, which varied sequences of normal and hot days, the function underestimates the fecundity reductions associated with more than one hot day interspersed with a single normal day (Fig. 5E).

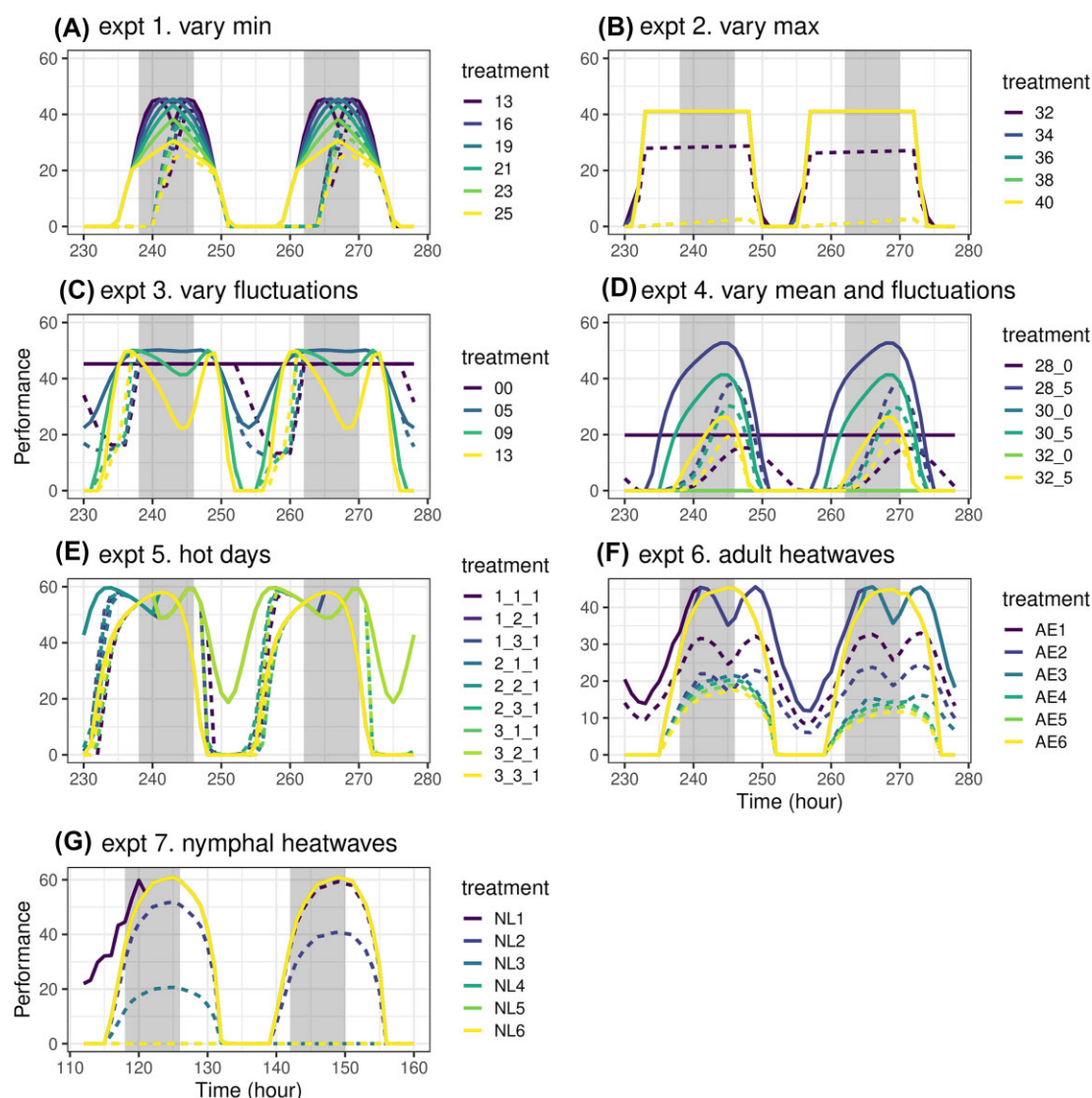


Fig. 4 The time dynamics of estimates of performance that determine fecundity (nymphs per adult) without (solid lines) and with (dashed lines, scenario 1) damage and repair vary among experiments and the treatments depicted in Fig. 2. Night hours are shaded. Parameter values correspond to fitting the two damage and two repair parameters (scenario 1).

We estimate a more pronounced exposure time than temperature effect (Table 2, Table S1). Dropping the time component impacts function fit more than dropping the temperature component, but the difference is small.

In Experiments 6 and 7, which implemented heatwaves during the adult and nymphal periods respectively, the function with damage and repair overestimates the fecundity declines resulting from long heat waves (Fig. 5F, G). Similarly to experiment 5, which also implemented heat waves, we estimate a stronger effect of exposure time than temperature (Table S1). Dropping the response to exposure time weakens fit to observations, whereas dropping the temperature effect has little consequence for fit. Both experiments estimate that the

onset of damage occurs at relatively low temperatures ($T_p = 0.84\text{--}0.85$), but scenario 2 weakens the fit to observations.

Development

Findings for how thermal stress influences development rate (Fig. 6, Table 2, Table S2) are qualitatively similar to those for fecundity, and so we focus our brief discussion on differences from findings for fecundity. Exposure time tends to be relatively more important for development time than for fecundity across scenarios (Table 2). We tend to estimate more pronounced damage and repair dynamics than observed (Fig. 6). However, our estimates sometimes miss pronounced declines in devel-

Table 2 Parameter estimates for the primary scenario 1 and the 7 experiments, along with SSE (sum of squared error) and AIC values for both fecundity (top) and development rate (bottom).

expt	d_{time}	d_{temp}	r_{mag}	r_{breadth}	c	T_r	scale	sse	AIC
Fecundity									
1	0.026079	0.000303	0.76	0.54	0.9	18	1	108	271
2	0.000113	0.002238	0.06	1.56	0.9	18	1	82	125
3	0.000661	0.000302	1	0.82	0.9	18	1.1	978	779
4	0.000030	0.000000	1	3	0.9	18	1.41	446	582
5	0.003822	0.000000	0.25	3	0.9	18	1.31	3497	2252
6	0.000001	0.005269	0.01	2.45	0.9	18	1	2101	1069
7	0.000742	0.000027	0.68	0.15	0.9	18	1.35	1463	987
Development rate									
1	0.000352	0.000000	0.1	2.97	0.9	25	1.36	0.0013	-689
2	0.486879	0.072730	1	2.57	0.9	25	1.21	0.0004	-203
3	0.000000	0.000000	1	0	0.9	25	1.29	0.005	-1255
4	0.496411	0.055453	1	2.68	0.9	25	1.62	0.007	-1012
5	0.030862	0.004284	0.5	3	0.9	25	1.49	0.0057	-3890
7	0.000779	0.470120	0.16	2.7	0.9	25	1.13	0.0015	-1809

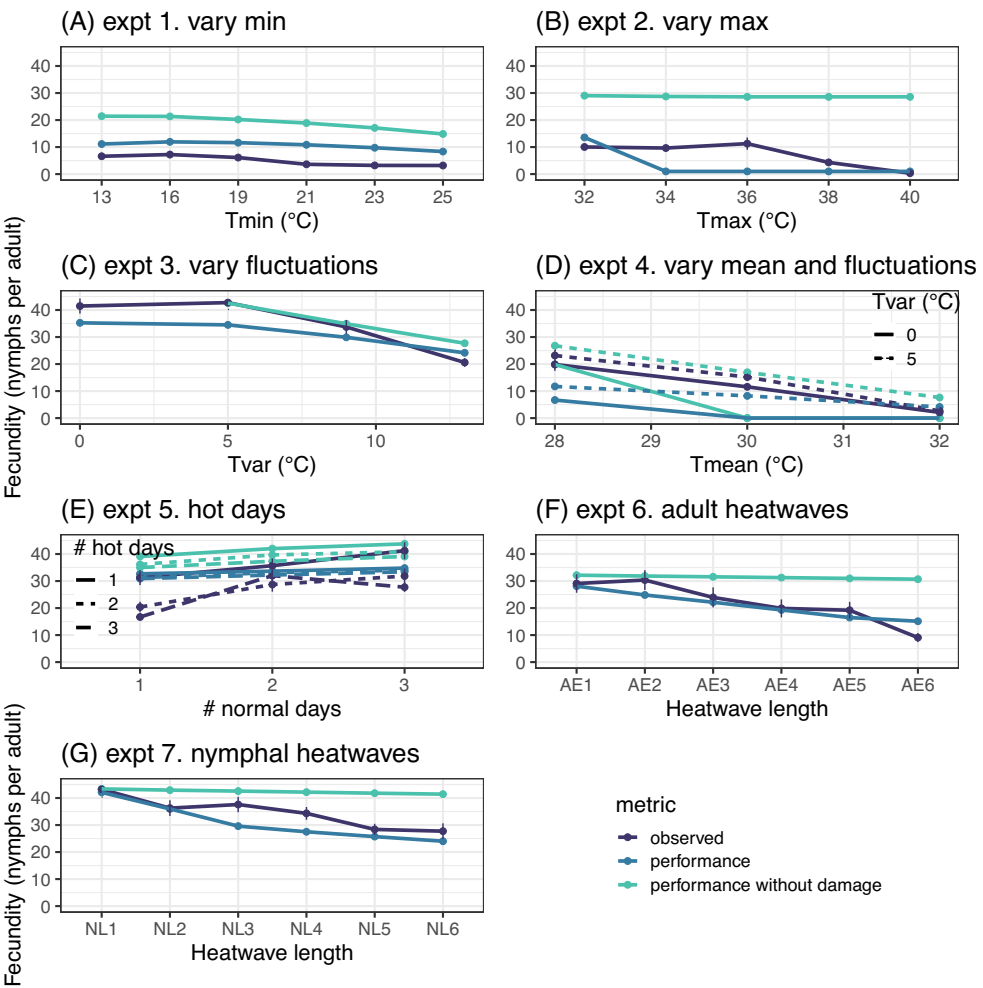


Fig. 5 Fecundity estimates tend to better correspond to observations (means $\pm$ SE) when they account for damage and repair. Parameter values correspond to fitting the two damage and two repair parameters (Table 2).

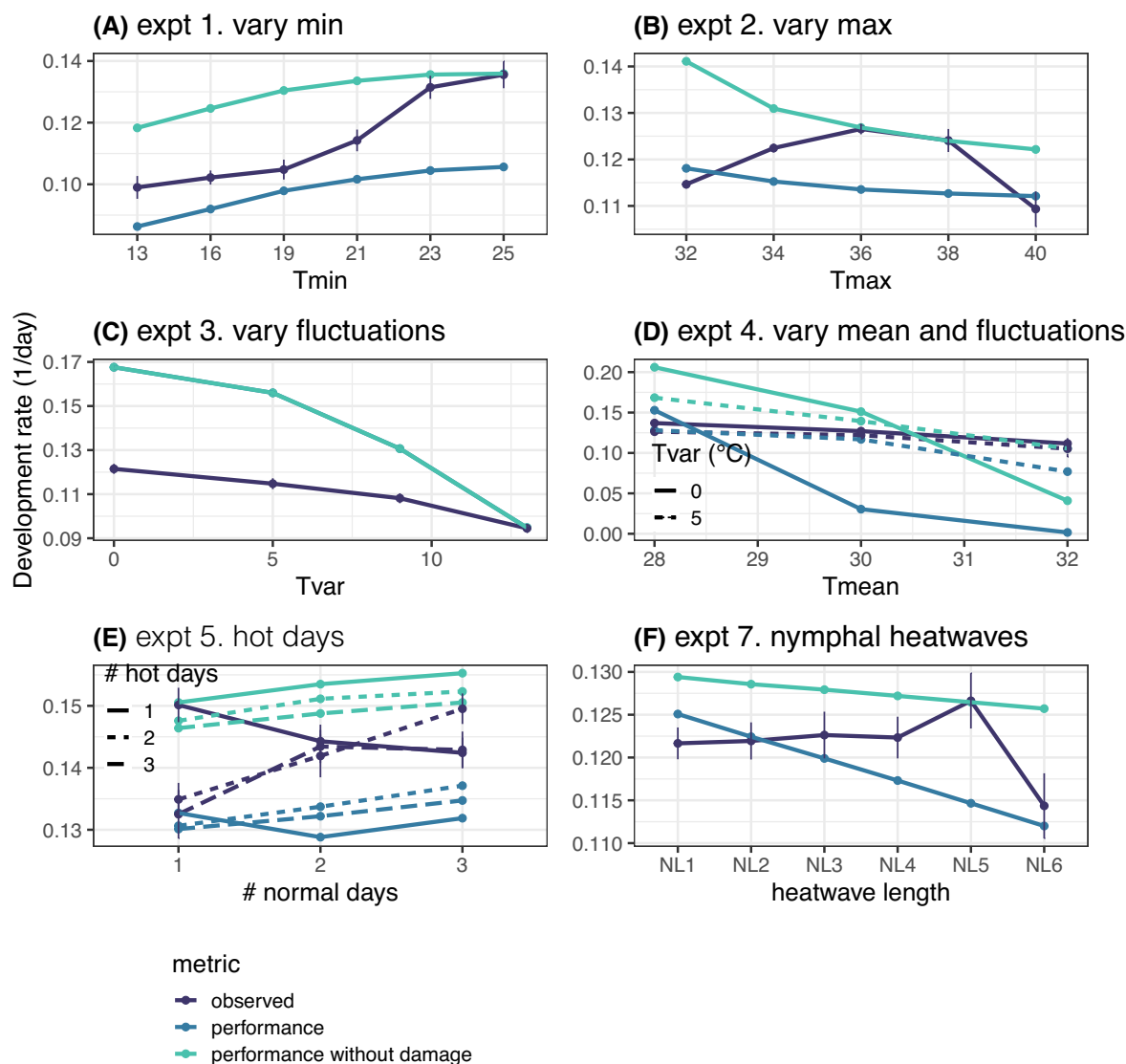


Fig. 6 The correspondence of development rate estimates either without or with damage and repair to observed values (means $\pm$ SE) varies among experiments. Parameter values correspond to fitting the two damage and two repair parameters (Table 1).

opment rate at low or high temperatures (expt 2). The estimates also miss some dynamics that are potentially consistent with acclimation (expt 5, 7).

Discussion

The aphid experiments collectively demonstrate how high temperatures above T_{opt} for prolonged periods, with limited opportunities for recovery, decrease fitness (Fig. 5). Our simple characterization of thermal stress and recovery captures some key dynamics uncovered in these experiments. The design of those experiments aids in distinguishing how the magnitude of diurnal fluctuations, how nighttime temperatures, and how variable durations of stressful and normal days influence damage and recovery. Our estimates align with

empirical findings (expt 1) that night warming (resulting in minimum temperatures exceeding T_{opt}) reduced aphid performance and survival by limiting opportunities for recovery (Zhao et al. 2014). Fitness components declined the most for Experiment 4, where we underestimate performance in thermal conditions with high means (Wang and Ma 2023). Uniquely for that experiment we analyzed a laboratory population. Field populations better tolerate high mean temperatures and temperature fluctuations but tend to exhibit lower performance in more mild conditions (Wang and Ma 2023). These findings point to the importance of better accounting for acclimation in response to environmental variability in considerations of thermal history.

We find that incorporating damage and repair in the model results in estimates of net performance over time that better match experimental observations (Fig. 5). Nevertheless, departures of estimates from observations for some experiments (summarized above) suggest limitations to our simple model of damage and repair. Some overestimates of the extent of acute damage likely result from using constant temperature TPCs, which incorporates chronic stress responses, as a baseline estimate of performance to which we add estimates of responses to acute stress. This highlights the need to refine TPC estimation approaches to better capture the timescales of responses to fluctuating body temperatures.

Roughly similar parameter estimates across some experiments provide insight into the dynamics of damage and repair (Table 2). However, similarly good fits are observed across scenarios dropping the time or temperature component of damage or omitting repair for some experiments (Table S1). Thus, while incorporating damage and repair dynamics improves fit to observed data (Fig. 5), our simple function is unable to consistently distinguish the forms of responses to thermal stress. This suggests that exposure time and temperature may similarly compound to stress organisms.

How does damage depend on the magnitude of thermal stress and respond to prolonged exposures?

Experiments varying maximum and minimum temperatures by night or day (expts 1 and 2) estimate similar responses to the magnitude and duration of exposure to stressful temperatures (d_{time} and d_{temp} respectively, Table 2). Experiments altering thermal variation (expts 3 and 4) detect stronger effects of the magnitude of thermal stress, whereas experiments varying the length of heatwaves (expts 5–7) primarily detect increasingly deleterious effects of the duration of exposure (Table 2). We are unable to resolve the shape of damage responses to exposure time and to temperature, but our approach is consistent with experimental and modelling findings that damage carries over and tends to accumulate additively (Ørsted et al. 2022).

The analysis indicates that the onset of acute damage occurs close to T_{max} (high c values in scenario 2; Table S1). This finding is contrary to prevailing thought that thermal stress initiates at temperatures well below critical thermal limits (Kingsolver and Woods 2016; Clusella-Trullas et al. 2021; Buckley et al. 2022). The finding may be an artifact of our using a TPC assay under constant temperatures, which may adequately (or over) account for stress responses until reaching T_{max} [beyond which TPCs rarely allow for negative performance, but see Kingsolver et al. (2013)].

What is the potential for recovery and repair and how does it depend on permissive temperatures?

The experiments estimate a substantial magnitude of potential repair (r_{mag}), except for experiment 2 and 6, which allow for prolonged recovery at near optimal temperatures. Estimates of the thermal breadth of repair (r_{breadth}) vary across experiments (Table 2). The experiments varying temperatures in consistent diurnal patterns (expts 1–4)—or those imposing periodic heatwaves (expt 5)—estimate that repair occurs over a broad temperature range (Table 2). Of these experiments, the narrowest repair breadth is estimated for experiment 1, which varies the daily minimum temperatures and thus opportunities for repair. We estimate narrow thermal ranges for repair for the experiments that impose a single heatwave of varying durations (expts 6–7, Table 2). We estimate both higher and lower than T_{opt} values for T_r (scenario 3, Table S1). We thus focus on the assumption that T_r equals T_{opt} . Although the capacity for repair is substantial, dropping repair from the model tends to drop AIC and have little impact on errors (scenario 6, Table S1). Contrary to an assumption that recovery and repair completes overnight (Rezende et al. 2020), our analyses indicate that damage persists and accumulates over extended time periods, such that overnight repair is incomplete (Colinet et al. 2015).

Conclusions

Our results build on previous research demonstrating that acute thermal stress increases with exposure time and the magnitude by which temperatures exceed a threshold for the onset of stress (Jørgensen et al. 2022). Other research, such as that refining the TDT framework, is making headway examining mechanistic response to acute thermal stress (Rezende et al. 2020; Ørsted et al. 2022, 2024). Our coarser temporal resolution analysis aims to investigate how thermal histories impact sub-lethal performance outcomes.

Our study also highlights dynamics of acclimation, damage, recovery, and repair that are missing from our estimates. Our analysis inadequately captures the fecundity reductions that occur when high nighttime temperatures (those exceeding T_{opt}) limit overnight recovery (expt 1). We underestimate the ability of aphids to maintain relatively stable fecundity up until high daily maxima temperatures (expt 2). We also underestimate the capacity of aphids to maintain fecundity through prolonged heatwaves (expt 6, 7). Both issues may stem from applying a constant temperature TPC to estimate performance in the absence of acute thermal stress. In contrast, we underestimate the decline in fecundity observed when multiple warm days are interspersed with

only one normal recovery day (expt 5). We tend to overestimate the influence of thermal stress on development rate (Fig. 6). These model limitations are consistent with acclimation (or the inability to acclimate when temperatures remain high).

Our analysis estimates parameters corresponding to rates of damage and repair by comparing performance aggregated across time series of environmental temperature to net experimental outcomes. Fitting parameters to more continuous assays of fitness rather than fitness outcomes at the end of the experiments would refine the parameterization and hopefully better capture the dynamics missing from our estimates. Systematically varying the duration and magnitude of thermally stressful conditions and monitoring the time course of recovery at multiple permissive, constant temperatures could better quantify damage and recovery.

Such experiments will also provide opportunities to refine TPC approaches to better account for timescales of exposure and sensitivity (Sinclair et al. 2016; Buckley et al. 2023). The consistency of fluctuations in the experiments limited our ability to assess responses to repeated exposures to acute periods of thermal stress. Comparing performance responses to constant, consistently fluctuating, and stochastically varying thermal conditions can inform TPC approaches (Kingsolver and Woods 2016). Comparing the temperature and timescale dependence of multiple fitness components including development, fecundity, and survival both empirically and through modeling will also clarify the role of thermal history (Buckley et al. 2021; Johnson et al. 2023). Testing the assumptions of and refining thermal tolerance landscapes will enable better accounting for thermal history (Rezende et al. 2014; Jørgensen et al. 2021). Establishing more continuity between studies of chronic performance responses to mild temperatures and studies of acute responses to stressful temperatures is needed.

The ability of organisms to behaviorally buffer thermal stress will shape fitness outcomes. Accounting for how behavior modifies body temperatures is a desirable first step in analyses of thermal history (Buckley et al. 2023). The implications of behavioral buffering can be complex as short-term stress avoidance can reduce selection to acclimate to or tolerate thermal stress, which can detriment fitness in the long-term as thermal refuges decline with warming (Buckley et al. 2015; Muñoz 2022). There are ample opportunities to examine how post-stress temperatures influence recovery (Ørsted et al. 2022) and whether animals select temperatures that maximize recovery rate. Accounting for thermal history is likely to be particularly important in terrestrial, temperate environments where environmental variation more frequently approaches thermal

limits (Vasseur et al. 2014; Buckley et al. 2022). Aquatic and sessile organisms with less potential for behavioral buffering may have greater capacity to mount acclimation responses to thermal variability (Gunderson and Stillman 2015).

Our analysis reinforces the need to account for stress responses that depend both on the magnitude and duration of exposure to stressful temperatures (Rezende et al. 2014; Jørgensen et al. 2022). Long exposure duration results in rapidly compounding damage because there is less chance for repair (expts 6 and 7). Our analysis also suggests that opportunities for recovery from thermal stress are contingent on permissive thermal conditions. Although our analysis is unable to resolve the details of responses to exposure time and temperatures, it does suggest that thermal histories of damage, recovery, and repair can substantially impact performance and fitness outcomes. Accounting for recent thermal history is likely to prove important to accurately anticipating the fitness implications of climate change and variability.

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Supplementary data

Supplementary data available at *ICB* online.

Conflict of interest

The authors have no conflicts of interest to declare associated with this manuscript.

Data availability

The data and code underlying this article are available in GitHub at <https://github.com/lbuckley/ThermalHistory> and in Zenodo at <https://zenodo.org/records/15484947>.

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